EXP 2

B

```
clc;
clear;
close all;

%% PART B: OPTIMUM FREQUENCY ANALYSIS

% Input Parameters
Z0 = 50;                    % Reference impedance
(ohms)
Zc = 52;                    % Characteristic
impedance (ohms)
L = 50e-3;                  % Transmission line
length (50 mm)
c = 3e8;                    % Speed of light (m/s)
er = 2.2;                   % Relative
permittivity
loss_dB_per_m = 10;         % Assumed attenuation
(dB/m)

% Frequency sweep from 8 GHz to 12 GHz
f = linspace(8e9, 12e9, 1001);

%% Calculate Transmission Line Parameters
beta = 2*pi*f*sqrt(er)/c;
alpha = loss_dB_per_m/8.686;
gamma = alpha + 1j*beta;

%% ABCD Parameters
A = cosh(gamma*L);
B = Zc*sinh(gamma*L);
C = sinh(gamma*L)/Zc;
D = cosh(gamma*L);

%% Calculate S-Parameters
den = A + B/Z0 + C*Z0 + D;

S11 = (A + B/Z0 - C*Z0 - D)./den;
S21 = 2./den;

%% Find Optimum Frequencies
[~, idx_min] = min(abs(S11));
[~, idx_max] = max(abs(S21));

f_best_S11 = f(idx_min);
f_best_S21 = f(idx_max);

%% Return Loss and Insertion Loss at Optimum S11
Frequency
RL_best = -20*log10(abs(S11(idx_min)));
IL_best = -20*log10(abs(S21(idx_min)));

%% Results at 10 GHz
[~, idx10] = min(abs(f - 10e9));

RL_10 = -20*log10(abs(S11(idx10)));
IL_10 = -20*log10(abs(S21(idx10)));

%% Display Results
fprintf('\nPART B: OPTIMUM FREQUENCY
ANALYSIS\n');
fprintf('================================\n');

fprintf('Minimum |S11| frequency = %.4f GHz\n',
...
    f_best_S11/1e9);

fprintf('Maximum |S21| frequency = %.4f GHz\n',
...
    f_best_S21/1e9);

fprintf('\nAt optimum S11 frequency:\n');
fprintf('|S11| = %.6f\n', abs(S11(idx_min)));
fprintf('|S21| = %.6f\n', abs(S21(idx_min)));
fprintf('Return Loss = %.4f dB\n', RL_best);
fprintf('Insertion Loss = %.4f dB\n', IL_best);

fprintf('\nAt 10 GHz:\n');
fprintf('Return Loss = %.4f dB\n', RL_10);
fprintf('Insertion Loss = %.4f dB\n', IL_10);

%% Plot 1: S11 versus Frequency
figure;
plot(f/1e9, 20*log10(abs(S11)), 'b', 'LineWidth',
1.5);
hold on;

plot(f_best_S11/1e9, ...
    20*log10(abs(S11(idx_min))), ...
    'ro', 'MarkerFaceColor', 'r');

xlabel('Frequency (GHz)');
ylabel('|S11| (dB)');
title('S11 versus Frequency');
legend('S11', 'Minimum |S11|');
grid on;

%% Plot 2: S21 versus Frequency
figure;
plot(f/1e9, 20*log10(abs(S21)), 'r', 'LineWidth',
1.5);
hold on;

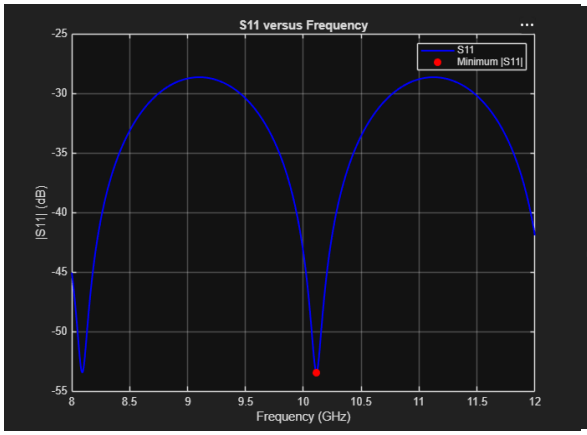
plot(f_best_S21/1e9, ...
    20*log10(abs(S21(idx_max))), ...
    'bo', 'MarkerFaceColor', 'b');

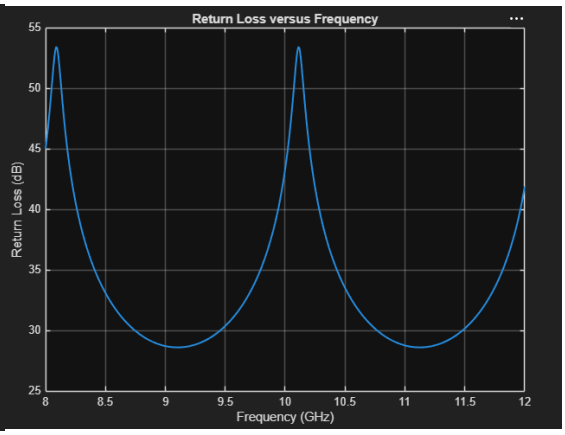
xlabel('Frequency (GHz)');
ylabel('|S21| (dB)');
title('S21 versus Frequency');
legend('S21', 'Maximum |S21|');
grid on;

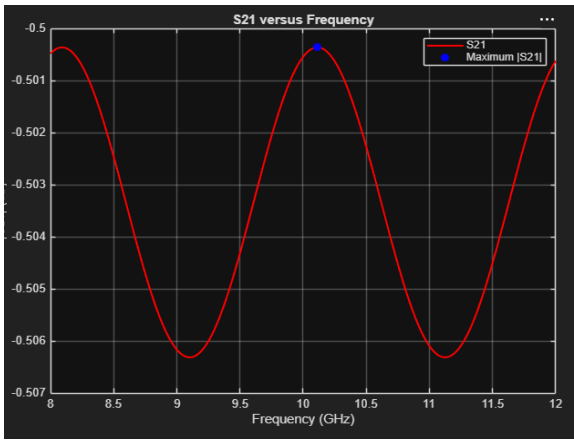
%% Plot 3: Return Loss versus Frequency
figure;
plot(f/1e9, -20*log10(abs(S11)), 'LineWidth',
1.5);
xlabel('Frequency (GHz)');
ylabel('Return Loss (dB)');
title('Return Loss versus Frequency');
grid on;

%% Plot 4: Insertion Loss versus Frequency
figure;
plot(f/1e9, -20*log10(abs(S21)), 'LineWidth',
1.5);
xlabel('Frequency (GHz)');
ylabel('Insertion Loss (dB)');
title('Insertion Loss versus Frequency');
grid on;

%% Compare Optimum Frequency with 10 GHz
fprintf('\nFREQUENCY COMPARISON\n');
fprintf('====================\n');
fprintf('Reference frequency = 10 GHz\n');
fprintf('Optimum S11 frequency = %.4f GHz\n', ...
    f_best_S11/1e9);
fprintf('Optimum S21 frequency = %.4f GHz\n', ...
    f_best_S21/1e9);
```



S11 versus Frequency
S11
Minimum |S11|
|S11| (dB)
Frequency (GHz)


Return Loss versus Frequency
Return Loss (dB)
Frequency (GHz)


S21 versus Frequency
S21
Maximum |S21|
Frequency (GHz)

